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Semiconductor device, integrated circuit and method of manufacturing the same

Abstract

A semiconductor device, an integrated circuit, and a method of manufacturing the same are provided. The semiconductor device includes a substrate. At least two thin-film transistors (TFT) are disposed over the substrate and electrically coupled to each other in parallel and a magnetoresistive random-access memory (MRAM) cell electrically couples to the thin-film transistors.

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Background/Summary

BACKGROUND

(1) Many electronic devices contain electronic memory, such as hard disk drives or random access memory (RAM). Electronic memory may be volatile memory or non-volatile memory. Non-volatile memory is able to retain its stored data in the absence of power, whereas volatile memory loses its data memory contents when power is lost. A magnetic random access memory (MRAM) device is currently explored to facilitate a static random access memory (SRAM) to own a high non-volatile storage density. The MRAM device includes an array of densely packed MRAM cells. In each MRAM cell, a magnetic tunneling junction (MTJ) element is integrated with a transistor to perform write and read operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a cross-sectional view of a memory device according to some embodiments of

the present disclosure.

(3) FIG. 2(a) illustrates a cross-sectional view of a memory device according to some embodiments of the present disclosure. FIG. 2(b) illustrates a top view of region A of the memory device illustrated in FIG. 2(a) according to some embodiments of the present disclosure.

(4) FIG. 3 illustrates a cross-sectional view of a memory device according to some embodiments of the present disclosure.

(5) FIG. 4 illustrates a cross-sectional view of an integrated circuit according to some embodiments of the present disclosure.

(6) FIG. 5A, FIG. 5B, FIG. 5C, FIG. 5D, FIG. 5E, FIG. 5F, FIG. 5G, and FIG. 5H illustrate a method of manufacturing a memory device such as the memory device of FIG. 2(a) according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

(7) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) As used herein, the terms such as “first,” “second,” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second,” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context.

(10) As used herein, the terms “approximately,” “substantially,” “substantial,” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation of less than or equal to +10% of that numerical value, such as less than or equal to +5%, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to +0.5%, less than or equal to +0.1%, or less than or equal to $\pm 0.05\%$. For example, two numerical values can be deemed to be “substantially” the same or equal if a difference between the values is less than or equal to $\pm 10\%$ of an average of the values, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to +0.5%, less than or equal to 0.1%, or less than or equal to $\pm 0.05\%$. For example, “substantially” parallel can refer to a range of angular variation relative to 0° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $+3^\circ$, less than or equal to $+2^\circ$, less than or equal to $\pm 1^\circ$, less than or

equal to $\pm 0.5^\circ$, less than or equal to $\pm 0.1^\circ$, or less than or equal to $\pm 0.05^\circ$. For example, “substantially” perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $\pm 3^\circ$, less than or equal to $\pm 2^\circ$, less than or equal to $\pm 1^\circ$, less than or equal to $\pm 0.5^\circ$, less than or equal to 0.1° , or less than or equal to $\pm 0.05^\circ$.

(11) An integrated circuit (IC) often comprises a back-end-of-line (BEOL) interconnect structure and semiconductor devices on a front side of a semiconductor substrate. The semiconductor devices may include, for example, various n-type metal-oxide (NMOS) and/or p-type metal-oxide semiconductor (PMOS) devices, such as transistors, capacitors, resistors, diodes, photo-diodes, fuses, magnetic random access memory (MRAM) cells, and the like. The semiconductor devices may, for example, be in the BEOL interconnect structure, and/or between the semiconductor substrate and the BEOL interconnect structure. A technical problem may reside in how to provide sufficient power to the semiconductor devices or how to reduce the power consumption thereof.

(12) In view of the foregoing, the present disclosure is directed to an IC, where semiconductor devices, such as MRAM cells are arranged within a BEOL interconnect structure, of which at least one of the technical problems mentioned above may be resolved.

(13) FIG. 1 illustrates a cross-sectional view of a memory device **100** according to some embodiments of the present disclosure. The memory device **100** includes a substrate **101**, a thin-film transistor (TFT) **140a**, and a magnetoresistive random-access memory (MRAM) cell **111**.

(14) The substrate **101** may be, for example, a bulk semiconductor substrate (e.g., a bulk silicon substrate) or a silicon-on-insulator (SOI) substrate, or a wafer. An SOI substrate may include a layer of a semiconductor material, such as silicon, formed on an insulator layer. The insulator layer may be, for example, a buried oxide layer or a silicon oxide layer. The insulator layer is provided on a substrate, typically, a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. The substrate **101** may include electrical devices such as various n-type metal-oxide (NMOS) and/or p-type metal-oxide semiconductor (PMOS) devices, such as transistors, capacitors, resistors, diodes, photo-diodes, fuses, and the like, interconnected to perform one or more functions. The functions may include memory structures, processing structures, sensors, amplifiers, power distribution, input/output circuitry, and the like.

(15) The process forming the individual devices such as transistors, capacitors, resistors, diodes, photo-diodes, fuses, and the like within the substrate **101** may be collectively referred as the front-end-of-line (FEOL) process, which is the first portion of integrated circuit (IC) fabrication where the individual devices (transistors, capacitors, resistors, etc.) are patterned in a substrate (e.g., wafer). FEOL generally covers everything up to (but not including) the deposition of metal layers.

(16) Following the FEOL process is the back-end-of-line (BEOL) process, which is the second portion of IC fabrication where the individual devices are interconnected with wiring or metal layers on the IC. The BEOL process generally begins when the first metal layer or M.sub.1 is deposited on the wafer. It may include contacts, insulating layers, metal layers, and bonding sites for chip-package connections. As a result, one or more metal layers, M.sub.1-M.sub.n may be formed over an interlayer dielectric (ILD) layer. A typical IC may include three or more metal layers, followed by a final passivation layer. The final passivation layer may be used for protecting the IC from mechanical abrasion during probe and packaging and to provide a barrier to contaminants. After the final passivation layer, the bonding pads for input/output will be formed, followed by a post-fabrication process such as wafer probe, die separation, and packaging. In more details, the BEOL process may include: adding a metal layer M.sub.n, adding an intra metal dielectric (IMD) layer, making vias through the IMD layer to connect to lower metal layer contacts, and forming higher metal layer contacts connected to the vias.

(17) The thin-film transistor (TFT) **140a** may be disposed over a first ILD layer **102** disposed over the substrate **101** during the BEOL process. A portion of the TFT **140a** may be disposed within the first ILD layer **102**. The first ILD layer **102** may include a low dielectric constant (k value less than

about 2.5) material. For example, the first ILD layer **102** may include, for example, an oxide, silicon dioxide (SiO₂), borophosphosilicate glass (BPSG), TEOS, spin-on glass (SOG), undoped silicate glass (USG), fluorinated silicate glass (FSG), high-density plasma (HDP) oxide, or plasma-enhanced TEOS (PETEOS).

(18) The TFT **140a** may include a gate electrode **103**, a gate dielectric layer **119**, and source/drain electrodes **104a**, **104b**. The TFT **140a** may be polycrystalline silicon TFT or amorphous silicon TFT.

(19) The gate electrode **103** is disclosed within the first ILD layer **102**. In some embodiments, the gate electrode **103** has an upper surface **103a** substantially coplanar with an upper surface **102a** of the first ILD layer **102**. The gate electrode **103** may include silicon, glass, plastic, or any other appropriate material, or may include a metal or any other appropriate conductive material. In some embodiments, the gate electrode **103** include a material selected from the group consisting of indium tin oxide (ITO), gallium zinc oxide (GZO), indium gallium zinc oxide (IGZO), indium gallium oxide (IGO), indium zinc oxide (IZO), indium oxide (In₂O₃), and a combination thereof. Also, a material used to form the gate electrode **103** may include a conductive metal selected from the group consisting of aluminum (Al), tungsten (W), copper (Cu), molybdenum (Mo), chromium (Cr), titanium (Ti), molybdenum tungsten (MoW), molybdenum titanium (MoTi), copper/molybdenum titanium (Cu/MoTi), and a combination thereof.

(20) The gate dielectric layer **119** is disposed over the first ILD layer **102**. The gate dielectric layer **119** may include a material selected from the group consisting of silicon oxide (SiO₂), silicon nitride (SiN_x), zirconium oxide (ZrO₂), hafnium oxide (HfO₂), titanium oxide (TiO₂), tantalum oxide (Ta₂O₅), a barium-strontium-titanium-oxygen compound (Ba—Sr—Ti—O), a bismuth-zinc-niobium-oxygen compound (Bi—Zn—Nb—O), and a combination thereof.

(21) The source/drain electrodes **104a**, **104b** are disposed over the gate dielectric layer **119**. The source/drain electrodes **104a**, **104b** are separated from each other by an active layer **107** formed over the gate dielectric layer **119** above the gate electrode **103**. The source/drain electrodes **104a**, **104b** may have an upper surface **113a**, **113b** substantially coplanar with an upper surface **107a** of the active layer **107**. The source/drain electrodes **104a**, **104b** may include the same material as that of the gate electrode **103**. For example, the source/drain electrodes **104a**, **104b** may include a metal or any other appropriate conductive material. In some embodiments, the source/drain electrodes **104a**, **104b** include a material selected from the group consisting of indium tin oxide (ITO), gallium zinc oxide (GZO), indium gallium zinc oxide (IGZO), indium gallium oxide (IGO), indium zinc oxide (IZO), indium oxide (In₂O₃), and a combination thereof. Also, a material used to form the source electrode **104a** and the drain electrode **104b** may include a conductive metal selected from the group consisting of aluminum (Al), tungsten (W), copper (Cu), molybdenum (Mo), chromium (Cr), titanium (Ti), molybdenum tungsten (MoW), molybdenum titanium (MoTi), copper/molybdenum titanium (Cu/MoTi), and a combination thereof.

(22) The active layer **107** serves to form a channel allowing electrons to move therein between the source/drain electrodes **104a**, **104b**. The active layer **107** may comprise an oxide semiconductor including, for example, silicon, IGZO, ITZO, IZO, AGZO, or a combination thereof.

(23) The MRAM cell **111** is electrically coupled to the TFT **140a**. The MRAM cell **111** may be electrically coupled to the TFT **140a** by a first metal line **112** disposed under the bottom electrode **114**. In some embodiments, the MRAM cell **111** is electrically coupled to the TFT **140a** through the first metal line **112**, the first interconnect line **106**, and the first conductive via **139**.

(24) A bit line (BL) may be electrically coupled to one end of the MRAM cell **111** by a second interconnect line **138** above the top electrode **120**. In some embodiments, the MRAM cell **111** is electrically coupled to the BL through the second conductive via **122**, the third conductive via **134**, and the second interconnect line **138**. A source line (SL) may be electrically coupled to an opposite end of the MRAM cell **111** through the TFT **140a**. Thus, application of a suitable word line (WL)

voltage to the gate electrode **103** of the TFT **140a** electrically couples the MRAM cell **111** between the BL and the SL. Consequently, by providing suitable bias conditions, the MRAM cell **111** can be switched between two states of electrical resistance, a first state with a low-resistance and a second state with a high-resistance, to store data. The MRAM cell **111** may be disposed within an interlayer dielectric (ILD) layer **128** over the substrate **101**. The MRAM cell **111** may include a bottom electrode **114**, a magnetic tunnel junction (MTJ) **116**, and a top electrode **120**.

(25) A portion of the bottom electrode **114** may be disposed in a dielectric layer **108** over the substrate **101**. The bottom electrode **114** may include, for example, tantalum, tantalum nitride, or ruthenium.

(26) The MTJ **116** may be disposed adjacent to the bottom electrode **114**. In some embodiments, the MTJ **116** is disclosed on the bottom electrode **114**. The MTJ **116** may include a lower ferromagnetic electrode **117** and an upper ferromagnetic electrode UFE, which may be separated from each other by a tunneling barrier layer **118**. In some embodiments, the lower ferromagnetic electrode **117** has a fixed or “pinned” magnetic orientation, while the upper ferromagnetic electrode UFE has a variable or “free” magnetic orientation, which may be switched between two or more distinct magnetic polarities that each represents a different data state, such as a different binary state. A “fixed” ferromagnetic layer refers to one having a magnetic orientation that is “fixed.” A “free” ferromagnetic layer refers to one that is capable of changing its magnetic orientation between two magnetic states. If the magnetic orientations of the pinned ferromagnetic electrode **117** and the free ferromagnetic electrode UFE are in a parallel orientation, it is more likely that electrons will tunnel through the tunneling barrier layer **118**, so the MTJ **116** is in a low-resistance state. Conversely, if the magnetic orientations of the pinned ferromagnetic electrode **117** and the free ferromagnetic electrode UFE are in an anti-parallel orientation, it is less likely that electrons will tunnel through the tunneling barrier layer **118**, so the MTJ **116** is in a high-resistance state. Because of this binary nature, the MTJ **116** may be used in memory cells to store digital data, with the low-resistance state corresponding to a first data state (e.g., logical “0”) and the high-resistance state corresponding to a second data state (e.g., logical “1”). In some embodiments, the MTJ may be vertically “flipped” such that the lower magnetic electrode **117** has a “free” magnetic orientation, while the upper ferromagnetic electrode UFE has a “pinned” magnetic orientation. In some embodiments, the lower ferromagnetic electrode **117** includes iron, cobalt, nickel, iron cobalt, nickel cobalt, cobalt iron boride, iron boride, iron platinum, iron palladium, or the like. In some embodiments, the upper ferromagnetic electrode UFE includes iron, cobalt, nickel, iron cobalt, nickel cobalt, cobalt iron boride, iron boride, iron platinum, iron palladium, or the like.

(27) The tunneling barrier layer **118** provides electrical isolation between the upper ferromagnetic electrode UFE and the lower ferromagnetic electrode **117**, while still allows electrons tunnel through under proper conditions. The tunneling barrier layer **118** may include, for example, magnesium oxide, aluminum oxide (e.g., Al.sub.2O.sub.3), nickel oxide, gadolinium oxide, tantalum oxide, molybdenum oxide, titanium oxide, tungsten oxide, or the like.

(28) The top electrode **120** electrically couples the second conductive via to the MTJ **116**. The top electrode **122** may include, for example, tantalum, tantalum nitride, or ruthenium.

(29) Compared to current non-volatile memory, such as flash random-access memory, MRAM typically is faster and has better endurance. Compared to current volatile memory, such as dynamic random-access memory (DRAM) and static random-access memory (SRAM), MRAM typically has similar performance and density, but lower power consumption. Therefore, MRAM is one promising candidate for next generation non-volatile electronic memory due to advantages over current electronic memory. Since MRAM is preferably disposed in the BEOL interconnect structure, and/or between the semiconductor substrate and the BEOL interconnect structure, a TFT is preferably used to provide power to the MRAM as the BEOL process cannot perform a temperature as higher as in the FEOL process (the process temperature in the BEOL process is typically below 400° C.), which makes a TFT a better choice compared to other transistors to be

electrically coupled to the MRAM as it can be made under the BEOL process temperature. Nevertheless, a disadvantage of this technology may be the low mobility and high contact resistance of the TFT.

(30) In view of the foregoing, the present disclosure further provides the following embodiments that may resolve at least one of the problems mentioned above.

(31) FIG. 2(a) illustrates a cross-sectional view of a memory device **200** according to some embodiments of the present disclosure. FIG. 2(b) illustrates a top view of region A of the memory device **200** illustrated in FIG. 2(a) according to some embodiments of the present disclosure. The memory device **200** illustrated in FIG. 2(a) is similar to that illustrated in FIG. 1 with a difference including that the MRAM cell **111** is electrically coupled to two TFTs **140a**, **240a** arranged in parallel and electrically connected to each other through a source/drain conductive via **243a**, **243b** and a gate conductive via **243c**. In some embodiments, the TFTs **140a**, **240a** are disposed between the MRAM cell **111** and the substrate **101** in a BEOL structure. In some embodiments, the MRAM cell **111** is disposed between the TFTs **140a**, **240a** and the substrate **101** in a BEOL structure.

(32) A second TFT **240a** is electrically coupled to the first TFT **140a**. The second TFT **240a** is electrically coupled to the first TFT **140a** in parallel so as to increase the current to the MRAM cell **111**. In some embodiments, the second TFT **240a** is disposed over the first TFT **140a**. In some embodiments, the second TFT **240a** and the first TFT **140a** are stacked.

(33) The source/drain conductive vias **243a**, **243b** electrically connect the first TFT **140a** to the second TFT **240a**. In some embodiments, the drain conductive via **243b** electrically connects the first TFT **140a** to the second TFT **240a** and the first interconnect line **106**, **244**. In some embodiments, the source conductive via **243a** electrically connects the first TFT **140a** to the second TFT **240a** and the first interconnect line **244**. In some embodiments, the source/drain conductive vias **243a**, **243b** extend from the source/drain electrodes **104a**, **104b** of the first TFT to the source/drain electrodes **204a**, **204b** of the second TFT and the first interconnect lines **244**, **106**. A source line (SL) may be electrically coupled to an opposite end of the MRAM cell **111** through the first TFT **140a** and the second TFT **240a** by the drain conductive via **243b**.

(34) The gate conductive via **243c** electrically connects the gate electrode **103** of the first TFT **140a** to the gate electrode **203** of the second TFT **240a**. In some embodiments, the gate conductive via **243c** extends from the gate electrode **103** of the first TFT **140a** to the gate electrode **203** of the second TFT **240a**. A suitable word line (WL) voltage to the gate electrode **103** of the first TFT **140a** electrically couples the MRAM cell **111** between the BL and the SL through the second TFT **240a**.

(35) By electrically connecting the first TFT **140a** and the second TFT **240a** in parallel by the drain conductive via **243b** to the first interconnect line **106** electrically connected to the MRAM cell **111**, the current flowing to the MRAM cell **111** may be increased. As a result, low current issue due to the low mobility and high contact resistance of a TFT transistor in a BEOL interconnect structure may be resolved and the MRAM cell **111** may thus receive sufficient power. In addition, the word line (WL) voltage that electrically couples the MRAM cell **111** between the BL and the SL may be reduced, which may in turn reduce the power consumption of the MRAM cell **111**, and thus the semiconductor device.

(36) The second TFT **240a** may include a gate electrode **203**, a gate dielectric layer **219**, and source/drain electrodes **204a**, **204b**. The TFT **240a** may be polycrystalline silicon TFT or amorphous silicon TFT.

(37) The gate electrode **203** is disclosed within an ILD layer **202**. The gate electrode **203** is similar to the gate electrode **103** of the first TFT **140a**. For example, the gate electrode **203** may include silicon, glass, plastic, or any other appropriate material, or may include a metal or any other appropriate conductive material. In some embodiments, the gate electrode **203** include a material selected from the group consisting of indium tin oxide (ITO), gallium zinc oxide (GZO), indium gallium zinc oxide (IGZO), indium gallium oxide (IGO), indium zinc oxide (IZO), indium oxide

(In.sub.2O.sub.3), and a combination thereof. Also, a material used to form the gate electrode **103** may include a conductive metal selected from the group consisting of aluminum (Al), tungsten (W), copper (Cu), molybdenum (Mo), chromium (Cr), titanium (Ti), molybdenum tungsten (MoW), molybdenum titanium (MoTi), copper/molybdenum titanium (Cu/MoTi), and a combination thereof.

(38) The gate dielectric layer **219** is disposed over the ILD layer **202** and the gate electrode **203**. The gate dielectric layer **219** is similar to the gate dielectric layer **119** of the first TFT **140a**. For example, the gate dielectric layer **219** may include a material selected from the group consisting of silicon oxide (SiO.sub.2), silicon nitride (SiN.sub.x), zirconium oxide (ZrO.sub.2), hafnium oxide (HfO.sub.2), titanium oxide (TiO.sub.2), tantalum oxide (Ta.sub.2O.sub.5), a barium-strontium-titanium-oxygen compound (Ba—Sr—Ti—O), a bismuth-zinc-niobium-oxygen compound (Bi—Zn—Nb—O), and a combination thereof.

(39) The source/drain electrodes **204a**, **204b** are disposed over the gate dielectric layer **219**. The source/drain electrodes **204a**, **204b** are separated from each other by an active layer formed over the gate dielectric layer **219** above the gate electrode **203**. The source/drain electrodes **204a**, **204b** are similar to those of the first TFT **140a**. For example, the source/drain electrodes **204a**, **204b** may have an upper surface substantially coplanar with an upper surface of the active layer. For example, the source/drain electrodes **204a**, **204b** may include a metal or any other appropriate conductive material. In some embodiments, the source/drain electrodes **204a**, **204b** include a material selected from the group consisting of indium tin oxide (ITO), gallium zinc oxide (GZO), indium gallium zinc oxide (IGZO), indium gallium oxide (IGO), indium zinc oxide (IZO), indium oxide (In.sub.2O.sub.3), and a combination thereof. Also, a material used to form the source electrode **204a** and the drain electrode **204b** may include a conductive metal selected from the group consisting of aluminum (Al), tungsten (W), copper (Cu), molybdenum (Mo), chromium (Cr), titanium (Ti), molybdenum tungsten (MoW), molybdenum titanium (MoTi), copper/molybdenum titanium (Cu/MoTi), and a combination thereof.

(40) The active layer of the second TFT **240a** is similar to those of the first TFT **140a**. For example, the active layer of the second TFT **240a** may comprise an oxide semiconductor including, for example, silicon, IGZO, ITZO, IZO, AGZO, or a combination thereof.

(41) FIG. 3 illustrates a cross-sectional view of a memory device **300** according to some embodiments of the present disclosure. The memory device **300** illustrated in FIG. 3 is similar to that illustrated in FIG. 2 with a difference including that the MRAM cell **111** is electrically coupled to more than two TFTs **140a**, **240a** arranged in parallel and electrically connected to each other through a source/drain conductive via **243a**, **243b** and a gate conductive via.

(42) By electrically connecting more TFTs to the MRAM cell **111** in parallel by the drain conductive via **243b**, the current flowing to the MRAM cell **111** may be further increased. As a result, low current issue due to the low mobility and high contact resistance of a TFT transistor in a BEOL interconnect structure may be resolved and the MRAM cell **111** may thus receive sufficient power. In addition, the word line (WL) voltage that electrically couples the MRAM cell **111** between the BL and the SL may be further reduced, which may in turn further reduce the power consumption of the device.

(43) FIG. 4 illustrates a cross-sectional view of an integrated circuit **400** according to some embodiments of the present disclosure. The integrated circuit **400** includes an embedded memory region **446a** comprising a MRAM cell **111** electrically connected to two TFTs **240a**, **140a** and a logic region **446b**.

(44) The embedded memory region **446a** is similar to that illustrated in FIG. 2 and is not described in details for brevity.

(45) In the logic region **446b**, a transistor **461** is disposed within the substrate **101** and the first ILD layer **455**. The transistor **461** may include a gate electrode **452**, a gate dielectric layer **454**, and source/drain regions **448a**, **448b**. An interconnect line **459** electrically connects to the transistor **461**

by a conductive via **457**.

(46) A second ILD layer **456**, a third ILD layer **462**, and a fourth ILD layer **468** may be disposed over the first ILD layer **455** sequentially, where each ILD layer **456**, **462**, **468** may include interconnect lines **460**, **466**, **472** and conductive vias **458**, **464**, **470** for electrically connecting the interconnect lines **460**, **466**, **472** to each other.

(47) FIGS. 5A-5H illustrate a method of manufacturing a memory device such as the memory device of FIG. 2(a).

(48) Referring to FIG. 5A, a first gate electrode **103** is disposed on a substrate **101** by performing a combination of a sputtering technology, a photolithography technology, and an etching technology with a conductive material. Subsequently, a first dielectric layer **102** having an upper surface **102a** coplanar with an upper surface **103a** of the first gate electrode **103** is deposited on the substrate **101** through a combination of a chemical vapor deposition (CVD) technology and a chemical-mechanical polishing (CMP) technology applied to a dielectric material.

(49) Referring to FIG. 5B, a first gate dielectric layer **119** is formed on the first gate electrode **103** and the first dielectric layer **102**. The first gate dielectric layer **119** may be formed by a chemical vapor deposition (CVD) technology or any other suitable technologies. Subsequently, an oxide semiconductor material **507** is deposited on the first gate dielectric layer **119**.

(50) Referring to FIG. 5C, a first active layer **107** is formed on the first gate dielectric layer **119** through a combination of a photolithography technology and an etching technology applied to the oxide semiconductor material **507**.

(51) Referring to FIG. 5D, first source/drain electrodes **104a**, **104b** are formed on the first gate dielectric layer **119** and separated by the first active layer **107** through a combination of a sputtering technology and a chemical-mechanical polishing (CMP) technology.

(52) Referring to FIG. 5E, an inter-level dielectric layer **242** is formed on the first active layer **107** and the first source/drain electrodes **104a**, **104b**. The inter-level dielectric layer **242** may be formed by a chemical vapor deposition (CVD) technology or any other suitable technologies.

(53) Referring to FIG. 5F, a second gate electrode **203** is disposed on the inter-level dielectric layer **242** by performing a combination of a sputtering technology, a photolithography technology, and an etching technology with a conductive material. Subsequently, a second dielectric layer **202** having an upper surface **202a** coplanar with an upper surface **203a** of the second gate electrode **203** is deposited on the inter-level dielectric layer **242** through a combination of a chemical vapor deposition (CVD) technology and a chemical-mechanical polishing (CMP) technology applied to a dielectric material.

(54) Referring to FIG. 5G, a second gate dielectric layer **219** is formed on the second gate electrode **203** and the second dielectric layer **202**. The second gate dielectric layer **219** may be formed by a chemical vapor deposition (CVD) technology or any other suitable technologies. Subsequently, a second active layer **207** is formed on the second gate dielectric layer **219** through a combination of a photolithography technology and an etching technology. Subsequently, second source/drain electrodes **204a**, **204b** are formed on the second gate dielectric layer **219** and separated by the second active layer **207** through a combination of a sputtering technology and a chemical-mechanical polishing (CMP) technology.

(55) Subsequently, a via **215a**, **215b** is disposed extending from the first source/drain electrodes **104a**, **104b** to the second source/drain electrodes **204a**, **204b**. The via **215a**, **215b** may be formed through a drilling technology (e.g., laser or mechanical drilling technology) or a combination of a photolithography technology and an etching technology.

(56) Referring to FIG. 5H, a conductive via **217a**, **217b** is formed by depositing a metal material into the via **215a**, **215b** through a combination of a sputtering technology and a chemical-mechanical polishing (CMP) technology. As a result, two TFTs electrically coupled to each other and arranged in parallel may be formed.

(57) The steps mentioned above may be repeated depending on the amount of the TFTs to be

formed. Afterwards, suitable technologies may be applied to dispose a MRAM cell electrically coupled to the TFTs. As a result, a memory device, such as the one illustrated in FIG. 2 may be obtained.

(58) In some embodiments, a semiconductor device is provided. The semiconductor device includes a substrate. At least two thin-film transistors (TFT) are disposed over the substrate and electrically coupled to each other in parallel and a magnetoresistive random-access memory (MRAM) cell are electrically coupled to the thin-film transistors.

(59) In some embodiments, an integrated circuit is provided. The integrated circuit includes a logic region and an embedded memory region. The logic region is disposed on a substrate. The embedded memory region includes at least two thin-film transistors (TFT) electrically coupled to each other in parallel and a magnetoresistive random-access memory (MRAM) cell is electrically coupled to the thin-film transistors.

(60) In some embodiments, a method of manufacturing a semiconductor device is provided. The method includes disposing at least two thin-film transistors (TFT) adjacent to a substrate, where the thin-film transistors are electrically coupled to each other in parallel; and disposing a magnetoresistive random-access memory (MRAM) cell over the thin-film transistors, where the MRAM cell is electrically coupled to the thin-film transistors.

(61) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor device, comprising: a substrate; at least two thin-film transistors (TFT) disposed over the substrate and electrically coupled to each other in parallel, wherein a gate dielectric layer is formed on a gate electrode of each of the thin-film transistors, and source/drain electrodes of each of the thin-film transistors are formed on the gate dielectric layer, and the two TFTs include a first TFT and a second TFT over the first TFT, and an inter-level dielectric layer, on which the gate electrode of the second TFT is formed, covers and is in contact with the source/drain electrodes and a channel between the source/drain electrodes of the first TFT; and a magnetoresistive random-access memory (MRAM) cell electrically coupled to the thin-film transistors, wherein the second TFT is positioned between the MRAM cell and the first TFT.
2. The semiconductor device of claim 1, wherein the source/drain electrodes of the second TFT is positioned between the MRAM cell and the gate electrode of the second TFT.
3. The semiconductor device of claim 1, wherein the source/drain electrodes of the first TFT are aligned with the source/drain electrodes of the second TFT.
4. The semiconductor device of claim 3, wherein the second TFT is electrically coupled to the first TFT by a first conductive via extending from the drain electrode of the first TFT to the drain electrode of the second TFT.
5. The semiconductor device of claim 4, wherein the gate electrode of the second TFT is electrically coupled to the gate electrode of the first TFT by a second conductive via extending from the gate electrode of the first TFT to the gate electrode of the second TFT.
6. The semiconductor device of claim 1, further comprising an interlayer dielectric (ILD) layer disposed over the substrate, wherein the first TFT is disposed over the ILD layer, and an upper surface of the ILD layer is substantially coplanar with an upper surface of the gate electrode of the

first TFT.

7. The semiconductor device of claim 1, further comprising an interlayer dielectric (ILD) layer disposed over the substrate, wherein the first TFT is disposed over the ILD layer, and an upper surface of an active layer of the first TFT is substantially coplanar with an upper surface of the source/drain electrodes of the first TFT.

8. The semiconductor device of claim 1, wherein the TFTs are disposed within a back-end-of-line (BEOL) interconnect structure.

9. An integrated circuit, comprising: a logic region disposed on a substrate; and an embedded memory region disposed over the substrate, comprising: at least two thin-film transistors (TFT) electrically coupled to each other in parallel, wherein a gate dielectric layer is formed on a gate electrode of each of the thin-film transistors, and source/drain electrodes of each of the thin-film transistors are formed on the gate dielectric layer, and the two TFTs include a first TFT and a second TFT over the first TFT, and an inter-level dielectric layer, on which the gate electrode of the second TFT is formed, covers and is in contact with the source/drain electrodes and a channel between the source/drain electrodes of the first TFT; and a magnetoresistive random-access memory (MRAM) cell electrically coupled to the thin-film transistors, wherein the second TFT is positioned between the MRAM cell and the first TFT.

10. The integrated circuit of claim 9, wherein the source/drain electrodes of the first TFT are aligned with the source/drain electrodes of the second TFT.

11. The integrated circuit of claim 10, wherein the second TFT is electrically coupled to the first TFT by a conductive via extending from the drain electrode of the first TFT to the drain electrode of the second TFT.

12. The integrated circuit of claim 11, wherein the gate electrode of the second TFT is electrically coupled to the gate electrode of the first TFT by a conductive via extending from the gate electrode of the first TFT to the gate electrode of the second TFT.

13. The integrated circuit of claim 9, further comprising an interlayer dielectric (ILD) layer disposed over the substrate, wherein the first TFT is disposed over the ILD layer, an upper surface of the ILD layer is substantially coplanar with an upper surface of the gate electrode of the first TFT, and an upper surface of an active layer of the first TFT is substantially coplanar with an upper surface of the source/drain electrodes of the first TFT.

14. The integrated circuit of claim 13, wherein the logic region comprises a transistor, and a portion of the transistor is within the ILD layer.

15. An integrated circuit, comprising: a substrate; an embedded memory region disposed over the substrate, comprising: a first thin-film transistor (TFT); a second TFT disposed over the first TFT and electrically coupled to the first TFT in parallel by a conductive via, wherein a gate dielectric layer is formed on a gate electrode of each of the first TFT and the second TFT, and source/drain electrodes of each of the first TFT and the second TFT are formed on the gate dielectric layer, and an inter-level dielectric layer, on which the gate electrode of the second TFT is formed, covers and is in contact with the source/drain electrodes and a channel between the source/drain electrodes of the first TFT; and a magnetoresistive random-access memory (MRAM) cell electrically coupled to the first TFT and the second TFT, wherein the second TFT is positioned between the MRAM cell and the first TFT.

16. The integrated circuit of claim 15, wherein the embedded memory region further comprises an interconnect line disposed over the first TFT and the second TFT, wherein the conductive via electrically connects the first TFT to the second TFT and the interconnect line.

17. The integrated circuit of claim 16, wherein the MRAM cell is disposed over the interconnect line and comprises a bottom electrode electrically connected to the interconnect line through a metal line.

18. The integrated circuit of claim 16, wherein the embedded memory region further comprises an interlayer dielectric (ILD) layer over the interconnect line, and the MRAM cell comprises a bottom

electrode partially in the ILD layer and electrically connected to the interconnect line.

19. The integrated circuit of claim 15, wherein the first TFT comprises a first drain electrode and a first source electrode at substantially a same elevation, and the second TFT comprises a second drain electrode vertically aligned with the first drain electrode and a second source electrode vertically aligned with the first source electrode.

20. The integrated circuit of claim 15, further comprising a logic region disposed on the substrate, wherein the logic region comprises a transistor disposed partially within the substrate.
